

Hongbo Li

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Education

Wayne State University

Ph.D. in Computer Graphics

Advisor: Zichun Zhong

Detroit, MI

2022-Present

University of Texas at Dallas

M.S. in Computer Science, Focus: Computer Graphics

Advisor: Xiaohu Guo

Dallas, TX

2018 – 2020

East China Normal University

B.S. in Software Engineering

Shanghai, China

2014 – 2018

Working Experience

Rendering Engineer

GritWorld GmbH.

May. 2020 - Jul. 2022

Shanghai, China

- Participated in refactoring the UI framework using Qt.
- Refined the Assimp converter to support more customized extensions used by the graphics engine, enabling saving of modified and newly generated data.
- Designed an upgrade workflow to support project versioning and developed migration tools to upgrade legacy assets and projects.
- Developed and integrated Progressive Mesh into the engine, increasing mesh facet support by up to 300% per scene.
- Utilized C++, OpenGL, GLSL, RenderDoc for development and debugging.

Publications

• NASM: Neural Anisotropic Surface Meshing

Hongbo Li, Haikuan Zhu, Sikai Zhong, Ningna Wang, Cheng Lin, Xiaohu Guo, Shiqing Xin, Wenping Wang, Jing Hua, Zichun Zhong

ACM SIGGRAPH Asia (Conference), 2024.

• Designing 3D Anisotropic Frame Fields with Odeco Tensors

Haikuan Zhu, Hongbo Li, Hsueh-Ti Derek Liu, Wenping Wang, Jing Hua, Zichun Zhong

ACM Transactions On Graphics (SIGGRAPH), 2025.

• ROSE: Multi-Level Super-Resolution-Oriented Semantic Embedding for Microvasculature Segmentation from Low-Resolution Images

Yifan Wang, Haikuan Zhu, Hongbo Li, Guoli Yan, Sagar Buch, Ying Wang, Ewart Mark Haacke, Jing Hua, Zichun Zhong

Neurocomputing, 2024.

• 3D Talking Face with Personalized Pose Dynamics

Chenxu Zhang, Saifeng Ni, Zhipeng Fan, Hongbo Li, Ming Zeng, Madhukar Budagavi, Xiaohu Guo

IEEE Transactions on Visualization and Computer Graphics (TVCG), 2021.

Technical Skills

Languages: Python, C++

Frameworks: Pytorch, TensorFlow, Geogram